

Mathematical and Computational Methods

Exercise Sheet 8: Differentiation II: applications

Tutorial route (►, in this order, about 80 minutes): 1.1, 2.4, 2.7, 2.6(a–d), 2.8, 2.10(a,c), 2.2(a,c). Everything else is for home study.

Levels: ★ drill, ★★ core, ★★★ challenge. Give exact answers unless a decimal is asked for.

1 Warm-up drills

Exercise 1.1 (► ★ drill, tangents & normals). For $f(x) = x^2$, find the equations of the tangent and the normal to the curve at the point $(2, 4)$.

Exercise 1.2 (★ drill, intermediate value theorem). Let $f(x) = x^3 + x - 1$.

(a) Show that $f(x) = 0$ has a solution in the interval $(0, 1)$.

(b) Show that this is the *only* real solution. (Hint: the sign of f' .)

Exercise 1.3 (★ drill, l'Hôpital). Evaluate, stating that the form is indeterminate before applying the rule:

$$(a) \lim_{x \rightarrow 0} \frac{\sin x}{x}, \quad (b) \lim_{x \rightarrow 0} \frac{e^x - 1}{x}, \quad (c) \lim_{x \rightarrow \infty} \frac{\ln x}{x^2}.$$

Exercise 1.4 (★ drill, classify). Locate and classify the stationary points of $f(x) = x^3 - 3x^2$ using the second-derivative test.

Exercise 1.5 (★ drill, asymptotes). Find the vertical and horizontal asymptotes of

$$(a) f(x) = \frac{1}{x-2}, \quad (b) g(x) = \frac{2x}{x-3}.$$

Exercise 1.6 (★ drill, extreme-value / closed interval). Use the closed-interval (extreme-value-theorem) method to find the absolute maximum and minimum of $f(x) = x^2 - 4x + 1$ on $[0, 3]$.

Exercise 1.7 (★ drill, optimisation — spot the error). Two non-negative numbers add to 10, and we want their largest possible product. A student writes the numbers as x and $10 - x$, sets $P(x) = x(10 - x)$, solves $P'(x) = 10 - 2x = 0$ to get $x = 5$, and concludes “ $P''(5) = -2 < 0$, so this is a minimum.” Identify the error, then find the largest product, checking the endpoints.

Exercise 1.8 (★ drill, gradient descent). Apply gradient descent to $f(x) = x^2$ with learning rate $\eta = 0.1$, starting from $x_0 = 1$. Compute x_1 and x_2 , and state the limit of the iteration.

2 Tutorial problems

Exercise 2.1 (★★ core, tangents & normals). Consider the curve $y = x^3 - 2x + 1$.

(a) Find the tangent and normal lines at the point where $x = 1$.

(b) Find the equations of *all* tangent lines to the curve that are parallel to the line $y = x$.

Exercise 2.2 (► ★★ core, *l'Hôpital — several forms*). Evaluate, first rewriting each into a $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form where necessary:

$$(a) \lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}, \quad (b) \lim_{x \rightarrow 0^+} x \ln x, \quad (c) \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right).$$

Exercise 2.3 (★★ core, *Newton's method*). The equation $x^3 - 2x - 5 = 0$ has a single real root, which the intermediate value theorem traps in $(2, 3)$ (since $f(2) = -1 < 0$ and $f(3) = 16 > 0$). Use *Newton's method* to pin it down: starting from $x_0 = 2$, write the iteration explicitly and carry out two steps by hand, tabulating x_n , $f(x_n)$ and $f'(x_n)$ at each stage to obtain x_1 and x_2 . Comment on how fast the estimate is settling.

Exercise 2.4 (► ★★ core, *classify — worked-example anchor*). Locate and classify all stationary points of

$$f(x) = 2x^3 - 3x^2 - 12x + 5.$$

Exercise 2.5 (★★ core, *full polynomial sketch*). Carry out a full curve-sketching analysis of $f(x) = x^3 - 6x^2 + 9x$ (domain/symmetry, intercepts, increase/decrease and extrema, concavity and inflection) and sketch the curve.

Exercise 2.6 (► ★★ core, *full rational sketch*). Let $f(x) = \frac{x}{x^2 - 1}$.

- State the domain and symmetry of f , its intercepts, and its vertical and horizontal asymptotes.
- Find $f'(x)$ and the intervals on which f increases or decreases. Does f have any stationary points?
- You may quote $f''(x) = \frac{2x(x^2 + 3)}{(x^2 - 1)^3}$. Find where f is concave up and concave down, and any point of inflection.
- Sketch the curve.

Exercise 2.7 (► ★★ core, *optimisation — worked-example anchor*). An open-topped box is made from a square sheet of card of side 12 cm by cutting a small square of side x from each corner and folding up the flaps. Find the value of x that maximises the volume, and the maximum volume.

Exercise 2.8 (► ★★ core, *physics: maximum power transfer*). A battery of EMF $\varepsilon \neq 0$ and internal resistance $r > 0$ delivers power $P(R) = \frac{\varepsilon^2 R}{(R + r)^2}$ to an external load of resistance $R > 0$. Show that the power delivered is greatest when $R = r$, and find that maximum power.

Exercise 2.9 (★★ core, *data science: least squares and the mean*). Given data values $x_1, \dots, x_n$, the sum of squared errors of a single predicted constant c is $L(c) = \sum_{i=1}^n (x_i - c)^2$.

- Show that L is minimised when c equals the mean $\bar{x} = \frac{1}{n} \sum_i x_i$.
- Find the minimising c for the data $\{2, 4, 9\}$.

Exercise 2.10 (► ★★ core, *gradient descent on a quadratic*). Minimise $f(x) = x^2 - 4x + 5$ by gradient descent with learning rate $\eta = 0.25$, starting from $x_0 = 0$.

- Write the update rule $x_{n+1} = g(x_n)$ explicitly and compute x_1, x_2, x_3 .
- To what value do the iterates converge, and why is that the right answer?
- What would go wrong if the learning rate were taken as $\eta = 1.5$?

3 Further practice (home)

Exercise 3.1 (★★core, mean value theorem).

- Find the value of c guaranteed by the mean value theorem for $f(x) = x^3$ on $[0, 2]$.
- Use the mean value theorem to prove that $|\sin a - \sin b| \leq |a - b|$ for all real a, b .

Exercise 3.2 (★★core, inverse-function theorem). Let $f(x) = 2x + \cos x$.

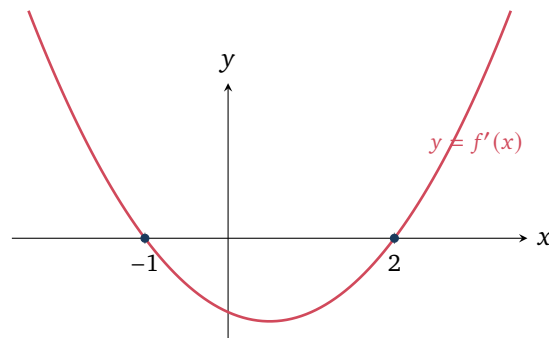
- Show that f is strictly increasing on $\mathbb{R}$, hence one-to-one and invertible.
- Compute $(f^{-1})'(1)$.

Exercise 3.3 (★★core, classify a mixed bag). Locate and classify all critical points of:

$$(a) f(x) = x^3 - 3x^2 + 3x, \quad (b) g(x) = x + \frac{1}{x}, \quad (c) h(x) = xe^{-x}, \quad (d) k(x) = x^{2/3}.$$

In (d), show first that $k'(0)$ does not exist.

Exercise 3.4 (★★core, interpret a graph of f'). The graph below shows the derivative f' of a function f (not f itself). Assume that outside the window f' continues as the picture suggests: it has no further zeros, it keeps falling to the left and it keeps rising to the right. Using only this information, describe f : state the intervals on which f is increasing or decreasing, locate and classify its stationary points, and say where f is concave up or down and where it has a point of inflection.



Exercise 3.5 (★★core, physics: kinematics min/max). A particle moves along a straight line with position $s(t) = t^3 - 9t^2 + 24t$ metres at time t seconds, for $0 \leq t \leq 5$. Find (a) the times at which it is instantaneously at rest, and (b) its maximum and minimum velocity on the interval.

Exercise 3.6 (★drill, true or false). For each statement, say whether it is true or false and justify briefly.

- If $f'(c) = 0$ then f has a local maximum or minimum at c .
- If $f''(c) = 0$ then c is an inflection point of f .
- A continuous function on an open interval need not attain a maximum value.
- If $f'(x) > 0$ on an interval, then f is strictly increasing there.
- At a point where $f'(a) = 0$, the normal line is horizontal.
- l'Hôpital's rule can be applied to any limit of a quotient.

Exercise 3.7 (★★★challenge, challenge: a chord property). Show that the tangent to $y = x^3$ at the point (a, a^3) (with $a \neq 0$) meets the curve again at exactly one further point, and that its x -coordinate is $-2a$.

Exercise 3.8 (★★★challenge, challenge: the Norman window). A “Norman” window is a rectangle topped by a semicircle, the semicircle’s diameter being the top of the rectangle. For a fixed perimeter P , find the proportions that maximise the area, and show that at the optimum the rectangle’s height equals the radius of the semicircle.